# Complex baseband representation

Corso di "Trasmissione dei segnali digitali"
Laurea Magistrale in Ingegneria Informatica

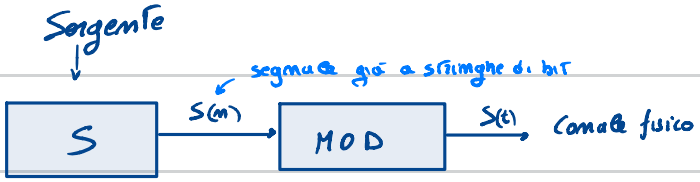
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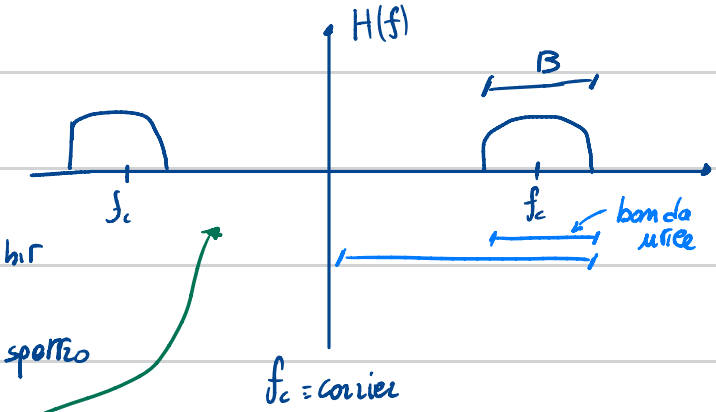

MODULATORE: associa una forma d'onda analogica a ogni simbolo o stringa di bit

↓

deve allocare questo segnale in una specifica porzione dello spettro delle frequenze, centrata attorno ad una frequenza portante $f_c$

$f_c$ = "CARRIER"

Il segnale poi attraversa il canale, che è caratterizzato da una sua risposta in frequenza e da una determinata banda a disposizione




Trasmettitore → porta nella specifica porzione dello spettro

↓ Per adattare il segnale rispetto alla capacità del canale

Ricevitore → porta in banda base

↓

Comporta che il segnale diventi complesso

→ In entrambi i momenti: Trasmissione e Ricezione

il segnale è complesso

Trasmissione  Complesso → Reale

Ricezione  Reale → Complesso

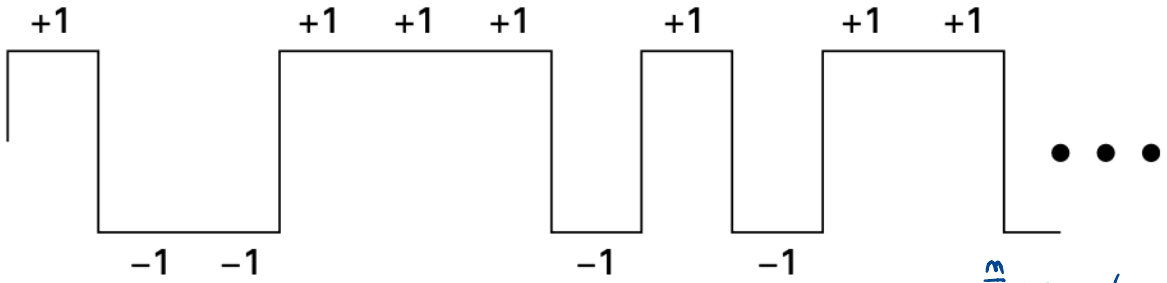
In entrambi i casi è un processo eseguito da un DSP

# Modulation

è il processo che permette di rappresentare l'informazione digitale tramite forme d'onda analogiche

- Modulation refers to the representation of digital information in terms of analog waveforms that can be transmitted over physical channels.

- A simple example of binary modulation is shown in figure:




I segnali di comunicazione devono essere allocati in certe porzioni dello spettro

$$s(t) = \sum_{i=1}^{m} b[m]\,\pi(t - mT)$$

$b = \{-1; 1\}$

- The symbols (bits) are mapped to an analog waveform by multiplying each symbol with translates of a rectangular pulse (*linear modulation*).

↳ perché $s(t)$ dipende linearmente dai simboli

- In the example, a time-limited transmit waveform is employed; the analog waveforms used for modulation are often constrained in the frequency domain, because of the physical characteristics of the communication medium or external factors (e.g., regulations on spectrum usage).

i segnali che devono essere trasmessi sul canale devono soddisfare vincoli nel dominio delle frequenze, che riguardano dove lo spettro del segnale sia allocato

# Baseband and passband signals (1/3)

- A signal $s_b(t)$ is said to be *baseband* if

$$S_b(f) \approx 0, \quad |f| > B \tag{1}$$

for some $B > 0$.

- A channel modeled as a linear time-invariant (LTI) system is said to be baseband if its transfer function $H(f)$ satisfies condition (1).

- A signal $s_p(t)$ is said to be *passband* if

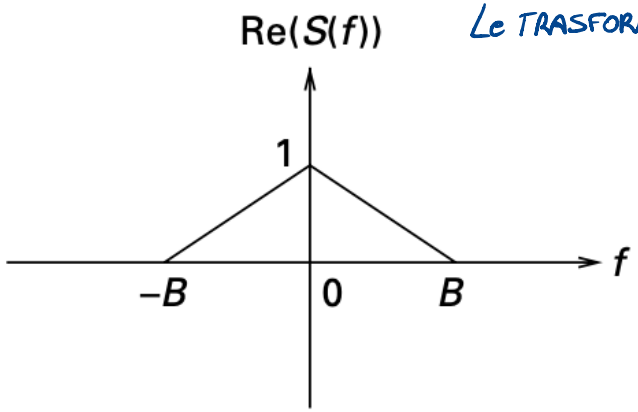
$$S_p(f) \approx 0, \quad |f \pm f_c| > B/2 \tag{2}$$

where $f_c > B > 0$ is referred to as "carrier" frequency.

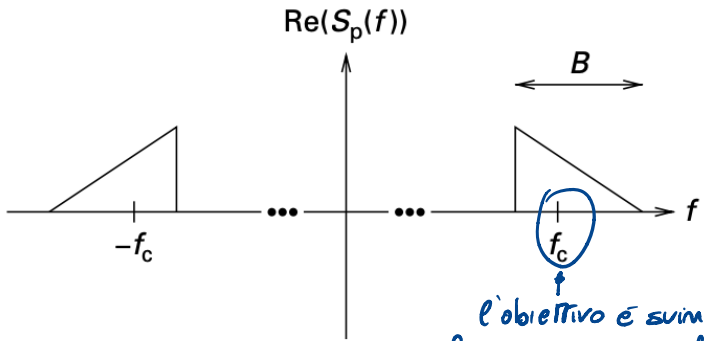
- A channel modeled as a LTI system is said to be passband if its transfer function $H(f)$ satisfies condition (2).

# Baseband and passband signals (2/3)

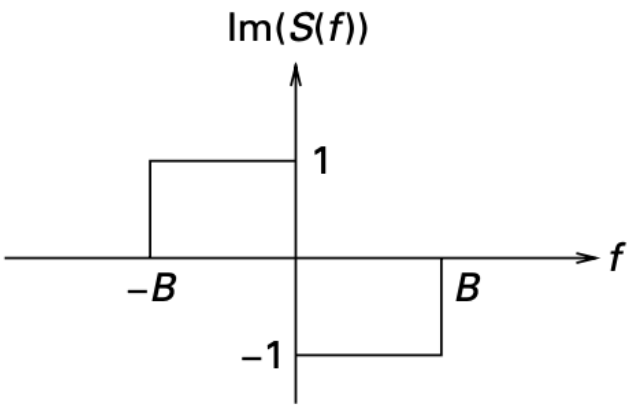
- Let $s_b(t)$ and $s_p(t)$ denote two real-valued baseband and passband signals, whose Fourier transforms (in either ordinary or generalized sense) are $S_b(f)$ and $S_p(f)$, respectively. Examples of these Fourier transforms are shown in the figure (we note that $S_b(f) = S_b^*(-f)$ and $S_p(f) = S_p^*(-f)$ for the conjugate symmetry property):

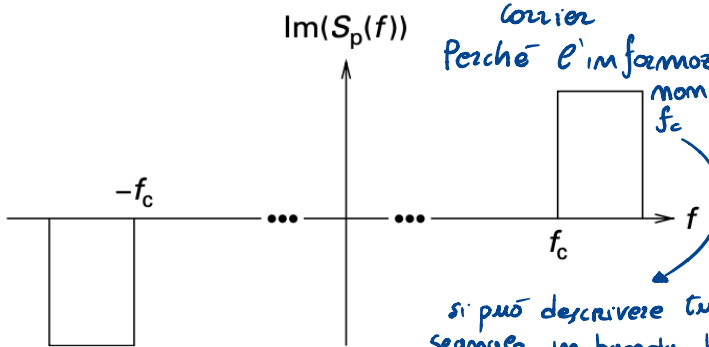



Le trasformate di Fourier dei segnali reali godono della simmetria hermitiana




l'obiettivo è svincolarci dalla $f_c$, ovvero dalla frequenza carrier




Perché l'informazione di $S_p(f)$ non dipende da $f_c$

si può descrivere tutto con un segnale in banda base

oltre al fatto che l'informazione non dipende da $f_c$,
Portare le segnale in banda base permette:

- Ridurre i costi hardware per il campionamento

- Modularità e indipendenza: posso riutilizzare componenti operanti nella banda base, senza costruirne ad hoc per la specifica banda del segnale
  ↳ Non posso generalizzare algoritmi e costa troppo calcolarla in digitale

# Baseband and passband signals (3/3)

- The bandwidth can be defined as the size of the frequency interval occupied by its Fourier transform.
- The term *one-sided* bandwidth refers to the spectral occupancy for the positive frequencies.
- The term *two-sided* bandwidth refers to the spectral occupancy for both positive and negative frequencies.
- For real-valued signals, the two-sided bandwidth is twice the one-sided bandwidth, because of the conjugate symmetry property $X(f) = X^*(-f)$.
- Usually, we will refer to the one-sided bandwidth simply as bandwidth.
- A passband signal is said to be *narrowband* if $B/f_c \ll 1$. From now on we will be dealing with narrowband passband signals.

Segnale a banda stretta è tale da avere una banda sia molto minore di $f_c$




$B \ll f_c$

Slide di sopra

Definizioni:

LARGHEZZA DI BANDA: dimensione dell'intervallo di frequenze occupato dalla trasformata di Fourier del segnale.

Si distingue tra banda MONOLATERA e BILATERA → considera sia le $f$ positive sia negative

se $s(t)$ è reale allora la BILATERA è il doppio della MONOLATERA

→ Simmetria Hermitiana $S(f) = S^*(-f)$

Segnale Passa banda a banda stretta:

un segnale passa banda, si dice a banda stretta se la sua B è molto minore della frequenza portante $f_c$

$$\frac{B}{f_c} \ll 1$$

Non è un ipotesi stringente

# Complex baseband representation (1/2)

- Passband channels are often employed and thus we must be able to transmit and receive passband signals.

- We will show that all the information carried in a real-valued passband signal is contained in a corresponding complex-valued baseband signal.

- This baseband signal is called the *complex baseband representation*, or *complex envelope*, of the passband signal.

- The complex envelope can be represented in discrete-time (i.e., sampled) at a much smaller sampling rate than the corresponding passband signal $s_p(t)$. As a consequence, modern communication transceivers can implement complicated signal processing algorithms, in a digital manner, directly on complex baseband signals, keeping the analog processing to the minimum.

# Complex baseband representation (2/2)

- The complex baseband representation leads to the following modular framework for transceiver design:
  - a digital signal processor (DSP) encodes information into the complex baseband waveform performing encoding, modulation and filtering;
  - the complex baseband waveform is *upconverted* to the corresponding passband signal to be sent on the physical channel;
  - the passband received waveform is *downconverted* to complex baseband at the receiver;
  - DSP performs operations on the complex baseband signal for synchronization, demodulation, and decoding.
- In this way, sophisticated algorithms are developed in complex baseband, independently of the physical frequency band used for communication.

# The analytical signal (1/3)

Per definire l'inviluppo dobbiamo costruire il segnale analitico $S_p^+(f)$ nel dominio f

- Let $s_p(t)$ denote a passband signal with Fourier transform $S_p(f)$, we define the **analytical signal** $S_p^+(f)$ associated with $S_p(f)$ as

$S_p^+(f)$ segnale analitico associato a $S_p(f)$ — annulliamo le frequenze negative (Ridondanti)

$$S_p^+(f) = \sqrt{2}\, S_p(f)\, \mathrm{u}(f) \tag{3}$$

fattore introdotto per motivi energetici → serve per conservare l'energia

where $\mathrm{u}(\cdot)$ is the unite step-function.

- Each passband signal is associated with an unique analytical signal and viceversa. Indeed:

$\sqrt{2}\, \mathcal{F}^{-1}[S_p(f)] * \mathcal{F}^{-1}[u(f)]$

$$s_p^+(t) \triangleq \mathcal{F}^{-1}[S_p^+(f)] = \sqrt{2}\, s_p(t) * \mathcal{F}^{-1}[\mathrm{u}(f)] \tag{4}$$

where $\mathcal{F}^{-1}[\cdot]$ is the inverse Fourier transform and $*$ denotes the convolution operator.

- We know that

$$\mathcal{F}[\mathrm{u}(t)] = \frac{1}{2}\,\delta(f) + \frac{1}{j2\pi f} \triangleq \mathrm{U}(f) \tag{5}$$

$$\mathcal{F}[\mathrm{U}(t)] = \mathrm{u}(-\mathrm{f}) \quad \text{(duality)} \tag{6}$$

# The analytical signal (2/3)

- From (5) and (6), it turns out that

$$\mathcal{F}^{-1}[\mathrm{u}(f)] = \mathrm{U}(-t) = \frac{1}{2}\,\delta(t) + \frac{j}{2\pi t} \tag{7}$$

where the reflection property of the Fourier transform has been applied.

- By substituting (7) into (4), it results that the analytical signal $s_p^+(t)$ associated with $s_p(t)$ can be expressed as

$$s_p^+(t) = \frac{1}{\sqrt{2}}\,s_p(t) + \frac{j}{\sqrt{2}}\,s_p(t) * \frac{1}{\pi t} = \frac{1}{\sqrt{2}}\,s_p(t) + \frac{j}{\sqrt{2}}\,\widehat{s_p}(t) \tag{8}$$

where

$$\widehat{s_p}(t) \triangleq s_p(t) * \frac{1}{\pi t} = \int_{-\infty}^{+\infty} \frac{s_p(\tau)}{\pi(t-\tau)}\,\mathrm{d}\tau \tag{9}$$

is the so-called Hilbert transform of $s_p(t)$.

# The analytical signal (3/3)

- From (8), it can be immediately inferred that

$$s_p(t) = \sqrt{2}\,\Re\{s_p^+(t)\} \tag{10}$$

with $\Re\{\cdot\}$ taking the real part. This concludes the proof.

- The analytical signal $s_p^+(t)$ admits the following Fourier transform:

$$S_p^+(f) \triangleq \mathcal{F}[s_p^+(t)] = \frac{1}{\sqrt{2}}\,S_p(f) + \frac{j}{\sqrt{2}}\,\mathcal{F}[\widehat{s_p}(t)] \tag{11}$$

where

$$\mathcal{F}[\widehat{s_p}(t)] \triangleq \widehat{S}_p(f) = S_p(f)\,\mathcal{F}\left[\frac{1}{\pi t}\right]. \tag{12}$$

- To calculate the Fourier transform in (12), we notice that $\mathrm{sgn}(t) = 2\,\mathrm{u}(t) - 1$; as a consequence, one has that $\mathcal{F}[\mathrm{sgn}(t)] = \frac{1}{j\pi f}$ and, thus, $\mathcal{F}\left[\frac{1}{\pi t}\right] = j\,\mathrm{sgn}(-f)$ (duality).

- Therefore, $\widehat{S}_p(f) = -j\,S_p(f)\,\mathrm{sgn}(f)$, with $\mathrm{sgn}(\cdot)$ denoting the signum function.

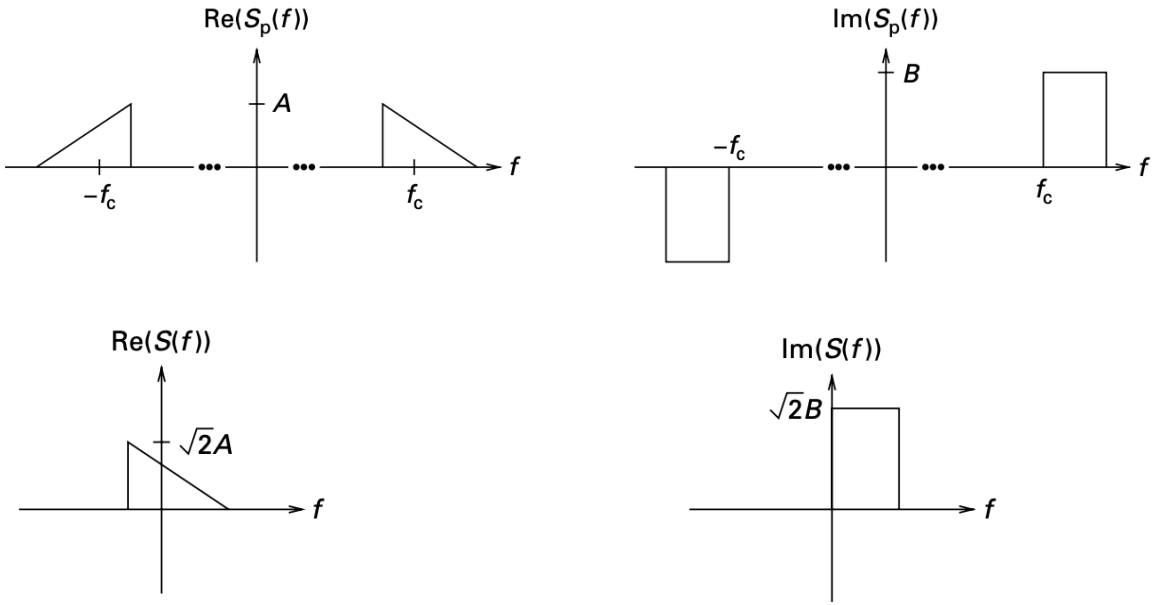
$$\mathcal{F}[\text{sign}(t)] = \mathcal{F}[2u(t) - 1] = 2\mathcal{F}[u(t)] - \delta(f) = \cancel{\delta(f)} + \frac{1}{j\pi f} - \cancel{\delta(f)} = +\frac{1}{j\pi f}$$

$$\mathcal{F}\left[\frac{1}{\pi t}\right] = j\,\text{sign}(f)$$

# Complex envelope

- The **complex envelope** is defined as

$$S(f) \triangleq S_p^+(f+f_c) = \sqrt{2}\, S_p(f+f_c)\, \mathrm{u}(f+f_c) \tag{13}$$




- Conversely, given the complex envelope $S(f)$ in the frequency domain, the passband signal is specified as (the proof can be easily derived from (16)):

$$S_p(f) = \frac{S(f-f_c) + S^*(-f-f_c)}{\sqrt{2}} \tag{14}$$

Parte Traslata da U(f)
← Relazione inversa

# Complex envelope in the time domain

- The time-domain expression of the complex envelope $s(t)$ associated with the passband signal $s_p(t)$ is given by

$$s(t) = \mathcal{F}^{-1}[S(f)] = \mathcal{F}^{-1}[S_p^+(f+f_c)] = s_p^+(t)\, e^{-j2\pi f_c t} = s_c(t) + j s_s(t) \qquad (15)$$

corrisponde ad una traslazione del segnale analitico
Proprietà di traslazione
Termini di seno e coseno

- The waveforms $s_c(t)$ and $s_s(t)$, referred to as the in-phase (or I) component and the quadrature (or Q) component of the passband signal $s_p(t)$, respectively, are real-valued signals.

$s_c(t)$ componente in-fase (o I)
$s_s(t)$ componente in QUADRATURA (o Q)
QUADRATURA indica che le prodotto ortogonale con $s_c$ è 0

- The relation between the passband signal $s_p(t)$ and its complex envelope $s(t)$ can be easily obtained as follows:

Abbiamo definito prima l'inviluppo come
$s(t) = s_p^+(t)\, e^{-j2\pi f_c t} \Rightarrow s_p^+(t) = s(t)\, e^{j2\pi f_c t} = [s_c(t) + j s_s(t)]\, e^{j2\pi f_c t}$

$$\begin{aligned} s_p(t) &= \sqrt{2}\,\Re\{s_p^+(t)\} = \sqrt{2}\,\Re\{s(t)\, e^{j2\pi f_c t}\} && (16) \\ &= \sqrt{2}\, s_c(t)\cos(2\pi f_c t) - \sqrt{2}\, s_s(t)\sin(2\pi f_c t) && (17) \end{aligned}$$

↳ in termini di seno e coseno

which is a valid representation for any passband signal $s_p(t)$.

# Energy of complex envelope

- The passband waveform $x_c(t) = \sqrt{2}\, s_c(t)\, \cos(2\pi f_c t)$ corresponding to the I component and the passband waveform $x_s(t) = \sqrt{2}\, s_s(t)\, \sin(2\pi f_c t)$ corresponding to the Q component are orthogonal, i.e.,

$$\langle x_c, x_s \rangle = \int_{-\infty}^{+\infty} x_c(t)\, x_s(t)\, \mathrm{d}t = 0 \quad \text{(for the proof refer to the textbook).} \tag{18}$$

- For real passband signals $u_p(t)$ and $v_p(t)$, with complex envelopes $u(t)$ and $v(t)$, respectively, the inner product satisfies

$$\langle u_p, v_p \rangle = \langle u_c, v_c \rangle + \langle u_s, v_s \rangle = \Re\{\langle u, v \rangle\} \tag{19}$$

- Specializing (19) to the inner product of a signal with itself, one can infer that the energy of the complex envelope is equal to that of the corresponding passband signal:

$$\|s_p\|^2 = \langle s_p, s_p \rangle = \Re\{\langle s, s \rangle\} = \Re\{\|s\|^2\} = \|s\|^2 . \tag{20}$$

$$\langle X_c, X_s \rangle = \int_{-\infty}^{+\infty} X_c(t) X_s(t)\, dt = \int_{-\infty}^{+\infty} X_c(f) X_s^*(f)\, df = \int_{-\infty}^{+\infty} S_c(f+f_c) + S_c(f-f_c) \left[ S_s^*(f+f_c) - S_s^*(f-f_c) \right] df \cdot \frac{1}{2j} =$$

Uguaglianza di Parseval

$$X_c = \frac{\sqrt{2}}{2} S_c(f) * [\delta(f-f_c) + \delta(f+f_c)]$$

$$X_s = -\frac{\sqrt{2}}{2j} S_s(f) * [\delta(f-f_c) - \delta(f+f_c)]$$

$$S_c(f+f_c) S_s^*(f+f_c) - S_c(f+f_c) S_s^*(f-f_c) + S_c(f-f_c) S_s^*(f+f_c) - S_c(f-f_c) S_s^*(f-f_c)$$

Per ipotesi i segnali sono a banda stretta, lo spettro centrato a $+f_c$ non si sovrappone mai a quello centrato in $-f_c$

perché $\frac{B}{f_c} \ll 1 \quad B \ll f_c$

$$= \frac{1}{2j} \left[ \int_{-\infty}^{+\infty} S_c(f-f_c) S_c^*(f-f_c) df - \int_{-\infty}^{+\infty} S_c(f+f_c) S^*(f+f_c)\, df \right] =$$

Sostituzione di variabile $f-f_c = f$ e $f+f_c = f$

$$= \frac{1}{2j} \left[ \underbrace{\int_{-\infty}^{+\infty} S_c(f) S_c^*(f) df - \int_{-\infty}^{+\infty} S_c(f) S^*(f)\, df}_{=0} \right] = 0$$

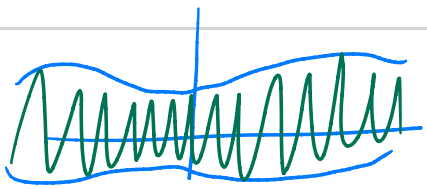
Verifica $\langle \mu_p, \nu_p \rangle = \mathrm{Re}\{\langle \mu, \nu \rangle\}$

$$\langle \mu_p, \nu_p \rangle = \int_{-\infty}^{+\infty} \mu_p(t)\, \nu_p(t)\, dt = 2 \int_{-\infty}^{+\infty} \mu_c(t) \nu_c(t) \underbrace{\cos^2(2\pi f_c t)}_{\frac{1+\cos(4\pi f_c t)}{2}} dt + 2 \int_{-\infty}^{+\infty} \mu_s(t) \nu_s(t) \underbrace{\sin(2\pi f_c t)}_{\frac{1-\cos(4\pi f_c t)}{2}} dt = \int_{-\infty}^{+\infty} \mu_c(t) \nu_c(t) + \mu_s(t) \nu_s(t) + \int_{-\infty}^{+\infty} \mu_c(t) \nu_c(t) \cos(4\pi f_c t) dt - \int_{-\infty}^{+\infty} \mu_s(t) \nu_s(t) \cos(4\pi f_c t)\, dt$$

$\simeq 0$ (Ammettiamo la componente in fase e in quadratura sono ortogonali)

$\simeq 0$

$\rightarrow$ Si annullano i termini misti



Le aree positive vengono bilanciate da quelle negative

Poiché $\frac{B}{f_c} \ll 1$

$$\simeq \mathrm{Re}\left\{\int_{-\infty}^{+\infty} \underbrace{[u_c(t)+ju_s(t)]}_{u(t)} \underbrace{[v_c(t)-jv_s(t)]}_{v^*(t)} dt\right\} = \mathrm{Re}\{\langle u, v\rangle\} \quad (19)$$

$u(t)$ ↑ inviluppo complesso di $u_p(t)$

Consideriamo

$$\langle s_p, s_p\rangle = \int_{-\infty}^{+\infty} s_p^2(t)\,dt = \|s_p\|^2 = \mathrm{Re}\{\langle s, s\rangle\} = \mathrm{Re}\left\{\int_{-\infty}^{+\infty} |s|^2\,dt\right\} = \|s\|^2$$

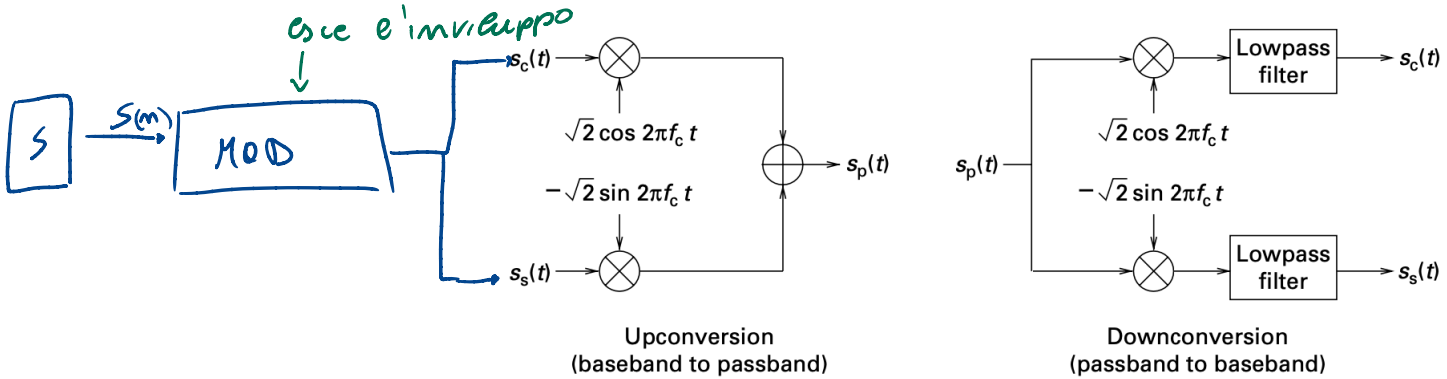
↑ è reale, quindi posso togliere Re

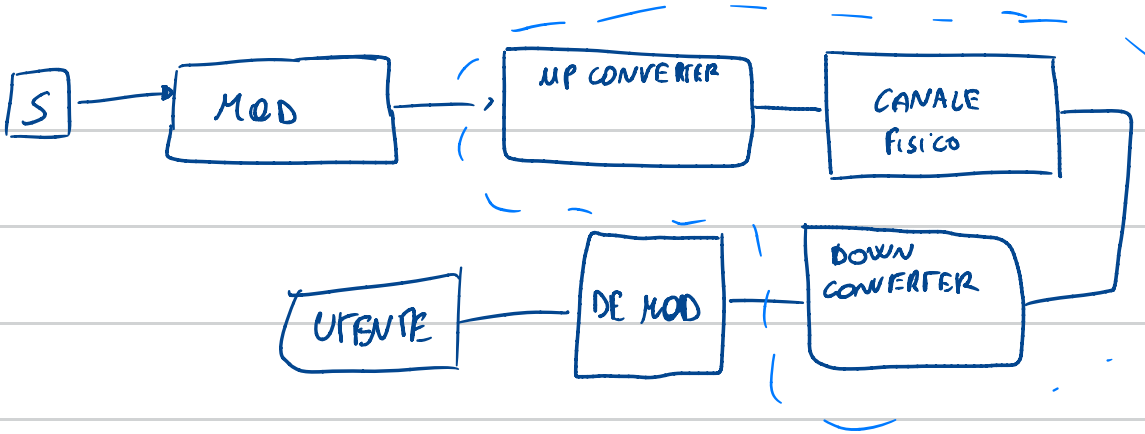
# Upconversion and downconversion (1/2)

- Equation (17) immediately tells us how to *upconvert* from baseband to passband.
- To *downconvert* from passband to baseband, we consider the product

$$\sqrt{2}\, s_p(t)\, \cos(2\pi f_c t) = 2\, s_c(t)\, \cos^2(2\pi\, f_c t) - 2\, s_s(t)\, \sin(2\pi\, f_c t)\, \cos(2\pi\, f_c t)$$
$$= s_c(t) + s_c(t)\, \cos(4\pi f_c t) - s_s(t)\, \sin(4\pi f_c t) \quad (21)$$

- Though lowpass filtering, the I component can be recovered. Similarly, we can obtain the Q component by lowpass filtering $-\sqrt{2}\, s_p(t)\, \sin(2\pi f_c t)$.






→ Modellato come un unico canale

dobbiamo capire se con tutti i passaggi che abbiamo fatto si conserva l'energia?

Si per le ortogonalità fatte $\sqrt{2}$

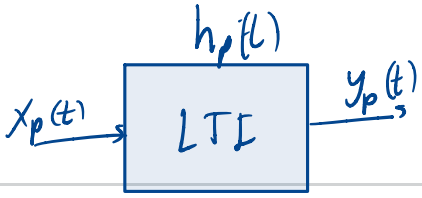
# Upconversion and downconversion (2/2)

- The complex baseband representation of the passband signal corresponds to deriving the "slow" amplitude and phase variations induced by baseband modulation, eliminating, instead, the rapid predictable phase variation due to the carrier frequency $f_c$.
- Since the phase variation due to $f_c$ is predictable, it does not convey any information.
- Information in a passband signal is entirely contained in its complex envelope.

# Complex baseband equivalent of passband filtering (1/3)

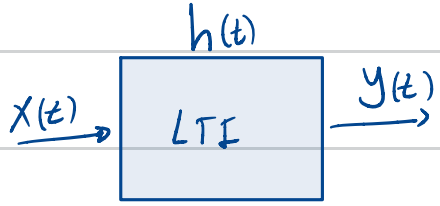
- Any passband filter can be implemented in complex baseband.
- Let $s_p(t)$ denote a passband signal which passes through a passband filter with impulse response $h_p(t)$ and let $y_p(t) = s_p(t) * h_p(t)$ be the *passband* filter output, with $*$ denoting the convolution operator.
- Let $s(t)$, $h(t)$, and $y(t)$ be the complex envelopes for $s_p(t)$, $h_p(t)$, and $y_p(t)$, respectively, with respect to a common frequency reference $f_c$, and $S(f)$, $H(f)$, and $Y(f)$ their corresponding frequency-domain counterparts.
- The input-output relationship of LTI filters in the frequency domain is $Y_p(f) = S_p(f)\, H_p(f)$. Focusing on the positive frequency segments $Y_p^+(f) = \sqrt{2}\, Y_p(f)\, \mathrm{u}(f)$, $S_p^+(f) = \sqrt{2}\, S_p(f)\, \mathrm{u}(f)$, and $H_p^+(f) = \sqrt{2}\, H_p(f)\, \mathrm{u}(f)$, one has

$$\begin{aligned} Y(f) = Y_p^+(f + f_c) &= \sqrt{2}\, Y_p(f + f_c)\, \mathrm{u}(f + f_c) \\ &= \sqrt{2}\, S_p(f + f_c)\, H_p(f + f_c)\, \mathrm{u}(f + f_c) = \frac{1}{\sqrt{2}} S(f)\, H(f) \end{aligned} \quad (22)$$




$$y_p(t) = x_p(t) * h_p(t)$$

↓ è uscita come diventa in termini di inviluppo complesso




Ragioniamo in f

$$Y_p(f) = X_p(f)\, H_p(f)$$

Segnale Analitico

$$\begin{cases} Y_p^+(f) = \sqrt{2}\, Y_p(f)\, u(f) \\ X_p^+(f) = \sqrt{2}\, X_p(f)\, u(f) \\ H_p^+(f) = \sqrt{2}\, H_p(f)\, u(f) \end{cases}$$

$$Y(f) = Y_p^+(f+f_c) = \sqrt{2}\, Y_p(f+f_c)\, u(f+f_c) = \sqrt{2}\, X_p(f+f_c) H_p(f+f_c) u^2(f+f_c) \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{1}{\sqrt{2}} X(f) H(f)$$

↑ Sostituisco e' uscita

Non altera nulla

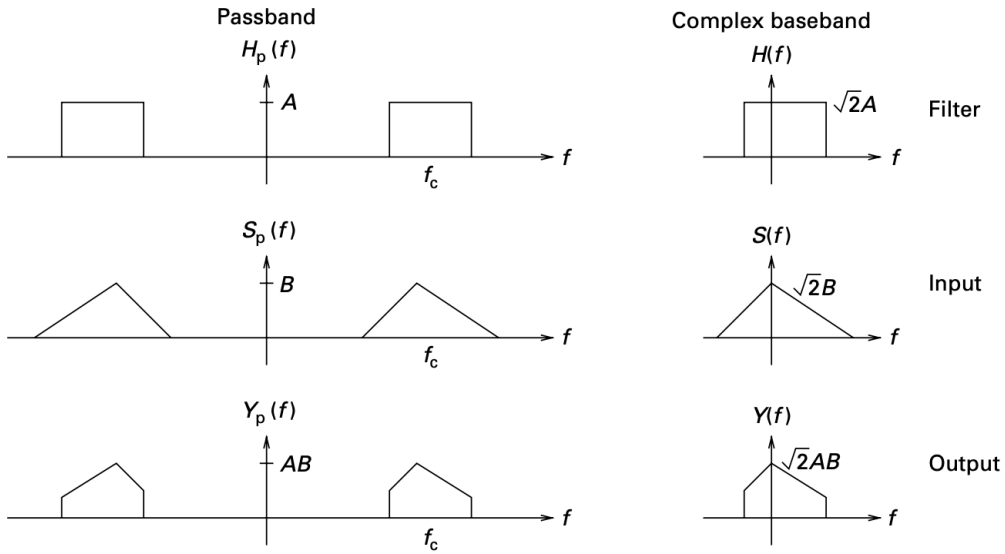
Resta fuori

L'inviluppo complesso di $y_p(t)$ nel dominio della frequenza $Y(f) = \frac{1}{\sqrt{2}} X(f) H(f)$

che si riduce nel dominio del tempo $y(t) = \frac{1}{\sqrt{2}} x(t) * h(t)$

# Complex baseband equivalent of passband filtering (2/3)

- The following figure depicts the relationship between the passband and complex baseband waveforms in the frequency domain:




- In the time domain one has

$$y(t) = \frac{1}{\sqrt{2}} s(t) * h(t) \tag{23}$$

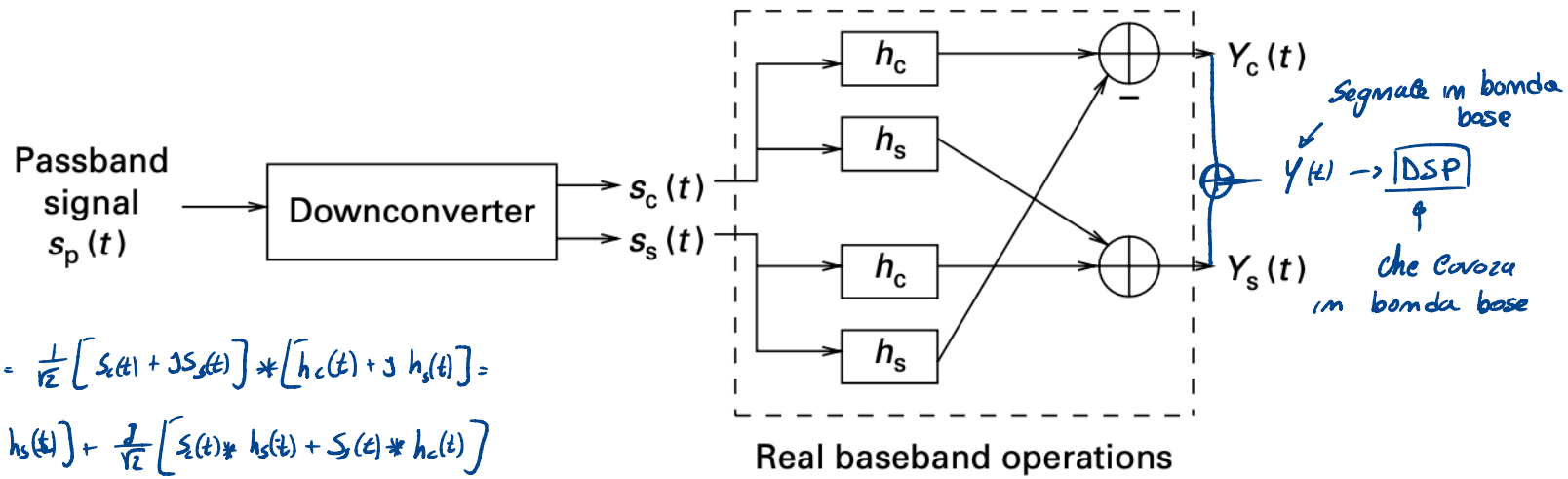
# Complex baseband equivalent of passband filtering (3/3)

- By expressing (23) in terms of I and Q components, we obtain

$$y_c(t) = \frac{1}{\sqrt{2}}[s_c(t) * h_c(t) - s_s(t) * h_s(t)] \quad (24)$$

$$y_s(t) = \frac{1}{\sqrt{2}}[s_s(t) * h_c(t) + s_c(t) * h_s(t)] \quad (25)$$

- The complex baseband realization of passband filter is depicted in the following figure:




$y(t) = \frac{1}{\sqrt{2}} s(t) * h(t) = \frac{1}{\sqrt{2}}\left[s_c(t) + j s_s(t)\right] * \left[h_c(t) + j\, h_s(t)\right] =$

$= \frac{1}{\sqrt{2}}\left[s_c(t) * h_c(t) - s_s(t) * h_s(t)\right] + \frac{j}{\sqrt{2}}\left[s_c(t) * h_s(t) + s_s(t) * h_c(t)\right]$

# Bandwidth

- We know that the spectral occupancy of a physical real-valued signal is the smallest band of positive frequencies that contains most of the signal content (for a real-valued signal $s(t)$ it results $S(-f) = S^*(f)$).
- For complex baseband signals, information is contained in both positive and negative frequencies.
- The spectral occupancy of a complex baseband signal is the smallest band around the origin that contains most of the signal content (including positive and negative frequencies).